Use Case 10: Infrastructure as Code

- **Scenario:** Automate infrastructure provisioning.
- **Tasks:**
  - Use CloudFormation to create a stack for a web application.
  - Include EC2, RDS, and an S3 bucket in the stack.
  - Update the stack to add a new feature.

Step 1: Define Requirements

1. The web application stack will include:
   - **EC2 instance**: For hosting the application.
   - **RDS instance**: For the database backend.
   - **S3 bucket**: For static file storage (e.g., images, documents).

2. The stack should be secure, cost-effective, and scalable.
3. Use **CloudFormation templates** to automate resource provisioning and ensure version control.

Step 2: Create a Basic CloudFormation Template

1. **Open CloudFormation Console**:
   - Navigate to the **AWS CloudFormation Console**.
2. **Write the Template**:
   - Use **YAML** or **JSON** to define your infrastructure.
3. **Save the Template**:
   - Save the file as web-app-stack.yaml.

Step 3: Deploy the CloudFormation Stack

1. **Upload the Template**:

- In the CloudFormation Console, click **Create Stack** and choose **With New Resources**.
- Upload the web-app-stack.yaml file.

2. **Specify Parameters**:
   - Enter required parameters like EC2 instance type and key pair.
3. **Review and Launch**:
   - Verify the resources and permissions.
   - Click **Create Stack**.
4. **Monitor the Stack Creation**:
   - Wait for the stack status to reach CREATE_COMPLETE.

Step 4: Secure and Optimize the Stack

1. **Secure the EC2 Instance**:
   - Restrict **SSH access** to specific IPs in the WebAppSecurityGroup.

- Enable **AWS Systems Manager Session Manager** for secure remote access without SSH:
  - Attach the AmazonSSMManagedInstanceCore policy to the EC2 instance's IAM role.

2. **Secure the RDS Instance**:
   - Use **KMS encryption** for the RDS storage.
   - Ensure the RDS instance is in a **private subnet** for better security.
3. **Secure the S3 Bucket**:
   - Block public access by enabling **Block Public Access Settings**.

1. **Cost Optimization**:
   - Use t3.micro instances for EC2 and RDS to stay within the **AWS Free Tier** (if applicable).

- Enable **Auto Scaling** for the EC2 instance to handle variable loads efficiently.
- Set up **lifecycle policies** to transition old S3 objects to **Glacier**.

Step 5: Update the Stack to Add a New Feature

1. **New Feature**:
   - Add an **Elastic Load Balancer (ELB)** to distribute traffic to multiple EC2 instances.
3. **Update the Stack**:
   - In the CloudFormation Console, click on your stack and choose **Update**.
   - Upload the modified template and deploy the update.
4. **Validate the Update**:

- Test the ELB endpoint to confirm traffic is properly distributed to EC2 instances.

Step 6: Monitor and Maintain the Stack

1. **Enable Logging**:
   - Use **CloudTrail** and **CloudWatch Logs** to track access and monitor resource usage.
   - Set up alarms for unusual activity.
2. **Automate Backups**:
   - Enable automated backups for RDS with a defined retention period.
   - Use **S3 Versioning** to preserve previous versions of objects.
3. **Optimize Costs**:
   - Use **Reserved Instances** or **Savings Plans** for predictable workloads.

- Periodically review and delete unused resources.